GUIDELINES ON NUS-IRB REVIEW NOT REQUIRED (RNR) DETERMINATION

1. In accordance with NUS' guidelines, all human subject research studies conducted by faculty and/or students must be subject to ethics review by either the NUS Institutional Review Board (IRB) or Departmental Ethics Review Committee (DERC). Further, all human biomedical research conducted in Singapore is governed by the Human Biomedical Research Act (HBRA) 2015, which mandates that any proposed human biomedical research should undergo review and be approved or exempted by an Institutional Review Board (IRB).

2. These guidelines delineate the ethical review of research involving:

   (i) **commercially available human biological materials (excluding human tissues and human stem cells) and cell lines (including modifications);** and/or

   (ii) **non-individually identifiable[1] human biological materials (excluding human tissues);** and/or

   (iii) **non-individually identifiable data (including non-individually identifiable health information).**

   - "Human biological material" is defined by the HBRA as "*any biological material obtained from the human body that consists of, or includes, human cells*".
   - "Human tissue" is defined by the HBRA as "*any human biological material but excludes human biological material specified in the First Schedule[1]*" (see Appendix 1).
   - "Health information" is defined by the HBRA as *"information pertaining to an individual (a) obtained in the course of or in connection with providing healthcare services; or (b) relating to the study, prevention, prognostication, diagnosis, or alleviation of a disease, disorder, or injury"*.

3. Certain provisions within the HBRA must be adhered to for research utilising human biological material that are included in the definition of "human tissue". These are:

   (i) Prohibition on commercial trading (i.e., buying and selling) of human tissue, as well as advertisements relating to such trading. However, this prohibition:
   - Does not apply to tissue products and derivatives, e.g. culture-expanded cell lines and tissue material that have undergone substantial manipulation and processing; and
   - Does not prevent payment to reimburse reasonable costs and expenses incurred in the process of collecting and supplying human tissue – including the removal of tissue from the donor, and the subsequent transportation, preparation, preservation, quality control, and storage of the tissue.

[1] **For the purposes of Review Not Required (RNR) category**: "***Non-individually identifiable***" means de-identified, unidentified, or anonymized data/samples, where the data/samples were obtained by the NUS researcher without any intervention or interaction with any research subject, and where the data/samples are not processed to generate any identifiable information or allow any individuals to be re-identified.

(ii) Appropriate consent from the donors must have been obtained for their tissues to be used for research, and the intended use is in accordance with any conditions or restrictions specified by the donors during consent. In the case where the human tissue has been imported from a place outside Singapore, appropriate consent must have been obtained in accordance with the legal or ethical requirements of that place. The IRB may request for demonstration/proof that appropriate consent was obtained from the donor(s) or from the place of origin for imported tissues.

4. **With effect from 25 March 2024,** NUS researchers will **no** longer need to apply to NUS-IRB for ethics review clearance, if his/her research study involves **solely**:

   (i) commercially available human biological materials (excluding human tissues and human stem cells) and cell lines (including modifications); and/or
   (ii) non-individually identifiable human biological materials (excluding human tissues); and/or
   (iii) non-individually identifiable data (including non-individually identifiable health information).

5. Studies qualifying for review not required (RNR) must meet **both** criteria as follows: -

   (i) The data or human biological material (excluding human tissues) intended for the RNR Study must be obtained by the researcher without any intervention or interaction with an individual person (e.g., research subject), and such data or human biological material (excluding human tissues) must be used, studied, or analysed in the RNR Study only; **and**

   (ii) The RNR Study should not obtain, use, study, analyse, or generate any identifiable information or identifiable human biological material (excluding human tissues), or enable any individual to be re-identified.

   For example: NUS researcher receives non-individually identifiable human biological materials from CIRB-approved research study for analysis. No identifiable information of donors will be provided to the NUS researcher.

6. This policy change is aligned with DSRB and CIRB's policies, based on the classification that a RNR Study which satisfies both criteria above, is not a human subject research study.

7. However, NUS researchers who wish to have NUS-IRB's written confirmation that review is not required for his/her study may submit his/her research application to the NUS-IRB via the iRIMS-IRB for a formal determination and issuance of a Review Not Required Determination Letter. Where applicable, NUS-IRB may request for supporting documents such as Material Transfer Agreement (MTA) or ethics approval letter(s) and consent form(s) from sample/ data sources, for review.

   Once the RNR Study is received and any outstanding queries on the research are successfully resolved, the IRB will issue a Review Not Required Determination Letter. The researcher can then commence the study.

8. However, please note that the following ***cannot*** undergo RNR:

   (i) Human stem cells and stem cell lines.
   (ii) Prohibited and Restricted human biomedical research as defined in the Third and Fourth Schedules of the HBRA 2015 (see Appendix 1);

(iii) Creation of a tissue bank or conduct of any tissue banking activities with the human biological materials / cell lines as defined in section 2 of the HBRA 2015 (see Appendix 1);

(iv) Use of human tissues (whether individually-identifiable or not);

(v) Use of individually-identifiable human biological materials;

(vi) Use of individually-identifiable data (including health information);

(vii) Use of any human biological materials to be injected into animals; or

(viii) Research requiring any type(s) of waiver of informed consent.

For the avoidance of doubt, point (i) applies to all:

- Embryonic stem cells and stem cell lines
- Fetal stem cells and stem cell lines
- Amniotic stem cells and stem cell lines
- Cancer stem cells and stem cell lines
- Adult stem cells and stem cell lines
- All forms of induced pluripotent stem cells and stem cell lines
- All types of stem cells that are totipotent, pluripotent, multipotent, oligopotent and unipotent.

For the avoidance of doubt, point (iv) applies to all:

- Primary tissues obtained directly from a human source without significant *in vitro* manipulation – but excludes malignant tissue microarrays which are commercially available;
- Patient samples that do not fall into the excluded categories in the First Schedule of the HBRA (see Appendix 1);
- Whole blood from a human source; and
- Human teeth.

9. **The IRB retains the right to make the final determination of whether a study requires ethics review clearance, subject to the applicable local laws and regulations.** See point one (page one) of this guideline. The IRB may reject or require a reclassification, or request for further information in order to facilitate review of the submitted applications.

10. Researchers with existing e-Declarations do not need to update the NUS-IRB via a modification submission on iRIMS-IRB should there be any changes to the list of human biological materials *(excluding human tissues and human stem cells)* and cell lines to be used in the research, unless he/she wishes to have NUS-IRB's written confirmation that review is not required for his/her modification (see paragraph 7 above).

11. Please contact the IRB Secretariat at 6516 4311 or email at irb@nus.edu.sg if you require further assistance or information.

Appendix 1 – Extracts from the Human Biomedical Research Act 2015

1) First Schedule of HBRA

Sections 2 and 62(1)

HUMAN BIOLOGICAL MATERIAL EXCLUDED FROM DEFINITION OF HUMAN TISSUE

1. Hair shaft, cut without dermal hair root or follicle.
2. Nail plate, cut without underlying dermal tissue.
3. Naturally excreted bodily fluids and waste products such as saliva, sweat, urine and faeces.
4. — (1) Any other human biological material that is not individually-identifiable and has been processed in such a manner that its functional, structural and biological characteristics are substantially manipulated as compared to the time of collection.

   (2) For the purposes of and without prejudice to the generality of sub-paragraph (1), human biological material is not deemed to be substantially manipulated merely because it has been processed by any of, or any combination of, the following methods:
   (a) cutting;
   (b) grinding;
   (c) shaping;
   (d) centrifugation;
   (e) soaking in antibiotic or antimicrobial solutions;
   (f) sterilization;
   (g) low-level irradiation;
   (h) cell separation, concentration or purification;
   (i) filtering;
   (j) lyophilisation;
   (k) freezing;
   (l) cryopreservation;
   (m) vitrification.

2) Third Schedule of HBRA

Sections 2, 30(1), and 62(1) and paragraph 3 of Fourth Schedule

PROHIBITED HUMAN BIOMEDICAL RESEARCH

1. Human biomedical research involving the development of human-animal combination embryos referred to in paragraph 2(a)(i) or (iii) of the Fourth Schedule beyond 14 days or the appearance of the primitive streak, whichever is the earlier.
2. Human biomedical research involving the implantation – –
   a. of a human-animal combination embryo mentioned in paragraph 2(a)(i) or (iii) of the Fourth Schedule into the uterus of an animal; or
   b. of a human-animal combination embryo into the uterus of a human.”; and
3. Human biomedical research involving the introduction of

human stem cells (including induced pluripotent stem cells) or human neural cells into the brain of living great apes whether prenatal or postnatal.

4. Human biomedical research involving the breeding of animals which have had any kind of pluripotent stem cells (including induced pluripotent stem cells) introduced into them.

3) Fourth Schedule of HBRA

Sections 2, 31(1) and 62(1) and paragraph 1 of Third Schedule

RESTRICTED HUMAN BIOMEDICAL RESEARCH

1. Human biomedical research involving human eggs or human embryos.
2. Human biomedical research involving —
   (a) the following types of human-animal combination embryos:
      (i) cytoplasmic hybrid embryos;
      (ii) human-animal combination embryos created by the incorporation of human stem cells (including induced pluripotent stem cells);
      (iii) human-animal combination embryos created in-vitro by using —
         (A) human gametes and animal gametes; or
         (B) one human pronucleus and one animal pronucleus;

   (b) the introduction of human stem cells (including induced pluripotent stem cells) into a prenatal animal foetus or animal embryo;

   (c) the introduction of human pluripotent stem cells (including induced pluripotent stem cells) into a living postnatal animal but excludes the introduction of such human pluripotent stem cells into immunodeficient mice solely for the analysis of teratoma induction;

   (d) the introduction of human stem cells (including induced pluripotent stem cells) or human neural cells into the brain of a living postnatal animal; or

   (e) any entity created as a result of any process referred to in sub-paragraphs (b), (c) and (d).
3. Nothing in this Schedule is to be construed to permit any human biomedical research that is prohibited under the Third Schedule.

4) Tissue Bank

Section 2

"Tissue Bank" means an individual or a body of persons, whether corporate or unincorporate, or other organisation, that carries on or conducts any tissue banking activity but excludes an individual, a body of persons or an organisation that conducts any tissue banking activity solely for the purpose of the person's or organisation's own human biomedical research approved or exempted from review by an institutional review board.

5) Tissue Banking Activity Section 2	"Tissue Banking Activity" means a structured and an organised activity involving human tissue for the purposes of facilitating current or future research or for public health or epidemiological purposes or any combination of such purposes including any of the following activities: (a) the collection, storage, procurement or importation of human tissue; (b) the supply, provision or export of human tissue.